House Price Prediction – Project Summary

1. Project Overview

This project focuses on predicting house prices using machine learning based on different property features. The workflow included data exploration, preprocessing, model building, evaluation, and visualization to identify the key factors influencing house prices.

2. Dataset Information

- Dataset: Housing Prices Dataset
- Records: 545
- Features: 13
- Target Variable: price

3. Data Preprocessing

The dataset was checked for missing values and duplicate records. Categorical features were converted into numerical form using One-Hot Encoding to prepare the data for model training.

4. Models Used

- Linear Regression
- Random Forest Regressor

5. Model Performance

Model	R^2 Score
Linear Regression	0.6279
Random Forest	0.6989

6. Key Findings

- "Area" was the most important feature affecting house prices.
- Bathrooms, parking, air conditioning, and stories also influenced price prediction.
- Random Forest outperformed Linear Regression.

7. Conclusion

This project successfully demonstrated the complete machine learning workflow for a regression problem. Among the two models, Random Forest achieved the best performance and provided more accurate house price predictions. The analysis also showed that both property size and amenities are important factors for estimating house prices accurately.